AI Engineer

Pre-requisites (One of these)
- Frontend
- Backend
- Full-Stack

Find the detailed version of this roadmap along with other similar roadmaps

roadmap.sh

Scrimba is offering 20% off to roadmap users on their AI Engineer course that covers this roadmap in depth. Check them out!

Scrimba - AI Engineer Path

Related Roadmaps
- ✓ AI and Data Scientist Roadmap
- ✓ Prompt Engineering
- ✓ Data Analyst Roadmap

Introduction

- Impact on Product Development
- Roles and Responsiblities
- What is an AI Engineer?
- AI Engineer vs ML Engineer

Common Terminology
- AI vs AGI
- LLMs
- Inference
- Training
- Embeddings
- Vector Databases
- AI Agents
- RAG
- Prompt Engineering

Using Pre-trained Models

Pre-trained Models
- Benefits of Pre-trained Models
- Limitations and Considerations
- OpenAI Models
  - Open AI Models
  - Capabilities / Context Length
  - Cut-off Dates / Knowledge

Popular AI Models
- Anthropic's Claude
- Google's Gemini
- Azure AI
- AWS Sagemaker
- Hugging Face Models
- Mistral AI
- Cohere
- Replicate

Open AI Platform

OpenAI API
- Chat Completions API
- Writing Prompts
- Open AI Playground
- Fine-tuning
- Prompt Engineering Roadmap

Managing Tokens
- Maximum Tokens
- Token Counting
- Pricing Considerations

AI Safety and Ethics

Understanding AI Safety Issues
- Prompt Injection Attacks
- Security and Privacy Concerns
- Bias and Fairness

Safety Best Practices
- OpenAI Moderation API
- Adding end-user IDs in prompts
- Conducting adversarial testing
- Robust prompt engineering
- Know your Customers / Usecases
- Constraining outputs and inputs

OpenSource AI
- Open vs Closed Source Models
- Popular Open Source Models
- Hugging Face

Finding Open Source Models
- Hugging Face Tasks
- Hugging Face Hub

Using Open Source Models
- Inference SDK
- Transformers.js

Ollama
- Ollama Models
- Ollama SDK

Embeddings & Vector Databases

What are Embeddings

Use Cases for Embeddings
- Semantic Search
- Data Classification
- Recommendation Systems
- Anomaly Detection

Open AI Embeddings API
- Open AI Embedding Models
- Pricing Considerations

Open-Source Embeddings
- Sentence Transformers
- Models on Hugging Face

Vector Databases
- Purpose and Functionality

Popular Vector DBs (pick one)
- Chroma
- Pinecone
- Weaviate
- FAISS
- LanceDB
- Qdrant
- Supabase
- MongoDB Atlas

Implementing Vector Search
- Indexing Embeddings
- Performing Similarity Search

RAG & Implementation
- RAG Usecases
- RAG vs Fine-tuning

Implementing RAG
- Chunking
- Embedding
- Vector Database
- Retrieval Process
- Generation

Ways of Implementing RAG
- Using SDKs Directly
- Langchain
- Llama Index

RAG Alternative
- Open AI Assistant API

AI Agents
- Agents Usecases
- Prompt Engineering
- ReAct Prompting

Building AI Agents
- Manual Implementation
- OpenAI Functions / Tools
- OpenAI Assistant API

Multimodal AI
- Multimodal AI Usecases

Multimodal AI Tasks
- Image Understanding
- Image Generation
- Video Understanding
- Audio Processing
- Text-to-Speech
- Speech-to-Text

Implementing Multimodal AI
- OpenAI Vision API
- DALL-E API
- Whisper API
- Hugging Face Models
- LangChain for Multimodal Apps
- LlamaIndex for Multimodal Apps

Development Tools
- AI Code Editors
- Code Completion Tools

Continue learning with following relevant tracks
- AI & Data Scientist
- Prompt Engineering